

Question No- I

$$\text{No of classes} = 9$$

$$\text{Range} = \text{Max value} - \text{Minimum value}$$

$$= 3462 - 3 = 3459$$

$$\text{width of class} = \frac{\text{Range}}{\text{No of classes}} = \frac{3459}{9} = 384.3333$$
$$\approx 384.5$$

(03) marks

Class Intervals

f

02 marks

$$3 \text{ under } 387.5 \quad \text{###, ###, ###, ###, ###, ###, ### = 33}$$

$$387.5 \quad \text{||} \quad 772 \quad \text{###, ###, |} = 11$$

$$772 \quad \text{||} \quad 1156.5 \quad \text{|||} = 3$$

$$1156.5 \quad \text{||} \quad 1541 \quad \text{||} = 2$$

$$1541 \quad \text{||} \quad 1925.5 \quad \text{0} = 0$$

$$1925.5 \quad \text{||} \quad 2310 \quad \text{0} = 0$$

$$2310 \quad \text{||} \quad 2694.5 \quad \text{|} = 1$$

$$2694.5 \quad \text{||} \quad 3079 \quad \text{0} = 0$$

$$3079 \quad \text{—} \quad 3463.5 \quad \text{|} = 1$$

02
marks

51

Question 2 NOI

(a) average Percentage increase 1999 - 2002 data.

$$G.M = (0.95 \times 1.06 \times 1.09 \times 1.04)^{\frac{1}{4}}$$
$$= (2.5132)^{\frac{1}{4}} = 1.03364$$

So the average rate of increase from 1999 - 2002 was 3.364 percent per year. (04) marks $\frac{1}{6}$

(b) $G.M = (0.95 \times 1.06 \times 1.09 \times 1.04 \times 1.07 \times 0.94)$

$$= 1.01741 \quad (04 \text{ marks})$$

So the new code appears to have slight effect on the rate of growth of convicts, which has decreased from 2 percent to 1.741 percent per year.

(Question 4)

Europe (bar chart)

02 marks

Asia (bar chart).

02 marks

or any other graph.

Question 4

Europe (arithmetic mean) = 34637 = $\bar{x}_1$ (02) marks

Asia (arithmetic mean) = 16326.333 = $\bar{x}_2$ (02) mark

x_i Europe	$(x_i - \bar{x}_1)^2$	Asia	$(x_i - \bar{x}_2)^2$
48704	197880489	26852	110789728.9
41441	46294416	23493	5136115.89
32679	3833764	18247	3688961.725
32136	6255001	13647	7178825.325
30384	18088009	9857	41852269.46
29326	28206721	5862	109502265.1
27789	46895104		

$$\bar{\sigma}_1 = \sqrt{\frac{\sum (x_i - \bar{x}_1)^2}{n}}$$

$$\bar{\sigma}_1 = 7045.297358$$

$$C.V = \frac{\bar{\sigma}_1}{\bar{x}_1} \times 100 \quad (02) \text{ marks}$$

$$= 20.34037$$

$$\bar{\sigma}_2 = \sqrt{\frac{\sum (x_i - \bar{x}_2)^2}{n}}$$

$$\bar{\sigma}_2 = 7352.69$$

$$C.V = \frac{\bar{\sigma}_2}{\bar{x}_2} \times 100 \quad (02) \text{ marks}$$

$$= 45.03583$$

Question 5

11/12/2021

C.I	C. f	f_i	x_i	$f_i x_i$	$f_i x_i - \bar{x}$
34 - 96	13	13	65	845	1605.24
97 - 159	15	2	128	256	120.96
160 - 222	15	0	191	0	0
223 - 285	20	5	254	1270	327.6
286 - 348	21	1	317	317	128.52
349 - 411	22	1	380	380	191.52
412 - 474	22	0	443	0	0
475 - 537	23	1	506	506	317.52
538 - 600	25	2	569	1138	761.04
				4712	3452.4

marks
03

$$Q_1 = l + \frac{h}{f} \left(\frac{n}{4} - c \right)$$

$$\left(\frac{n}{4} \right)^{th} = \frac{25}{4} = 6.25^{th} \text{ observation}$$

$$34 - 96 \rightarrow Q_1 \text{ group}$$

$$= 34 + \frac{62}{13} (6.25 - 0)$$

$$= 63.8076$$

(03)
marks

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{4712}{25} = 188.48$$

$$Q_3 = l + \frac{h}{f} \left(\frac{3n}{4} - c \right)$$

$$\frac{3n}{4} = 3 \times 6.25 = 18.75$$

$$223 - 285 \rightarrow Q_2$$

$$Q_3 = 223 + \frac{62}{5} (18.75 - 15)$$

$$= 269.5$$

(03)
marks

$$= 188.48 \quad (02) \text{ marks}$$

$$\text{mean absolute deviation} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$$

$$= \frac{3452.4}{25} = 138.096$$

(02) marks

(b) $f_i (x_i - \bar{x})^2$

$$198215.0352$$

$$7315.6608$$

$$0$$

$$21464.352$$

$$16517.3904$$

$$36679.9104$$

$$0$$

$$100818.9504$$

$$289590.9408$$

$$\sum f_i (x_i - \bar{x})^2$$

= minimum

(04) marks

$f_i (x_i - Q_1)$

$$1$$

$$1$$

$$0$$

$$1$$

$$5$$

$$2$$

$$\sum f_i (x_i - Q_1)^2$$

(03)

marks

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